

Análise Nodal – Lista de Exercícios 1*

RESOLUÇÃO

3.2 Para o circuito da Figura 3.51, obtenha v_1 e v_2 .

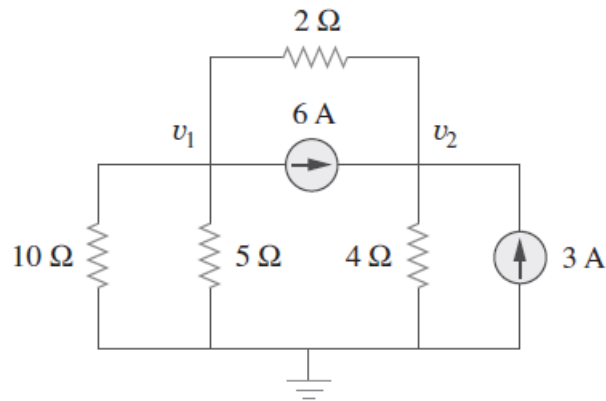
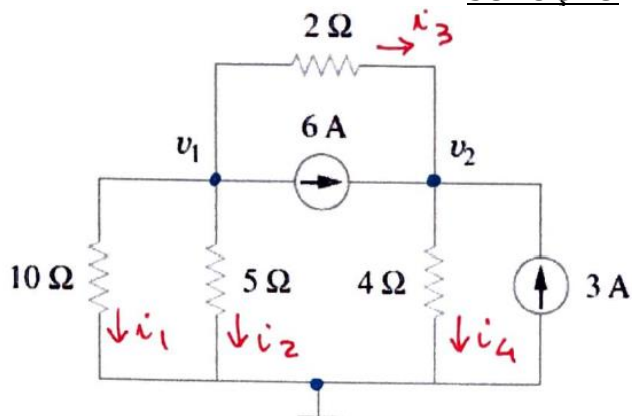


Figura 3.51 Esquema para o Problema 3.2.

SOLUÇÃO



no 1:

$$i_1 + i_2 + i_3 + 6 = 0$$

$$\frac{v_1}{10} + \frac{v_1}{5} + \frac{v_1 - v_2}{2} + 6 = 0$$

$$\Rightarrow v_1 + 2v_1 + 5v_1 - 5v_2 + 60 = 0$$

$$\Rightarrow \boxed{8v_1 - 5v_2 = -60}$$

no' 2:

$$i_3 + 6 + 3 = i_4$$

$$\frac{v_1 - v_2}{2} + 9 = \frac{v_2}{4}$$

$$2v_1 - 2v_2 + 36 = v_2$$

$$\Rightarrow \boxed{2v_1 - 3v_2 = -36}$$

$$\begin{cases} 8v_1 - 5v_2 = -60 \\ 2v_1 - 3v_2 = -36 \end{cases} \xrightarrow{\times (-4)} -8v_1 + 12v_2 = 144$$

$$\Rightarrow 7v_2 = 84 \quad \Rightarrow \boxed{v_2 = 12V}$$

$$\boxed{v_1 = 0}$$

3.3 Determine as correntes I_1 a I_4 e a tensão v_o no circuito da Figura 3.52.

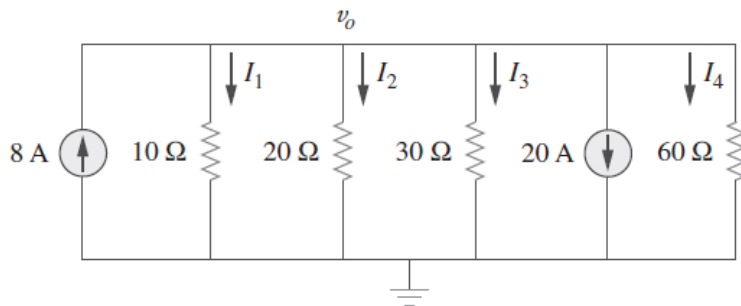


Figura 3.52 Esquema para o Problema 3.3.

SOLUÇÃO

$$I_1 + I_2 + I_3 + I_4 + 20 = 8$$

$$\frac{v_o}{10} + \frac{v_o}{20} + \frac{v_o}{30} + \frac{v_o}{60} = -12$$

$$6v_o + 3v_o + 2v_o + v_o = -720$$

$$12v_o = -720$$

$$\Rightarrow \boxed{v_o = -60 \text{ V}}$$

$$\Rightarrow \boxed{I_1 = -6 \text{ A}}$$

$$\boxed{I_2 = -3 \text{ A}}$$

$$\boxed{I_3 = -2 \text{ A}}$$

$$\boxed{I_4 = -1 \text{ A}}$$

3.4 Dado o circuito da Figura 3.53, calcule as correntes i_1 a i_4 .

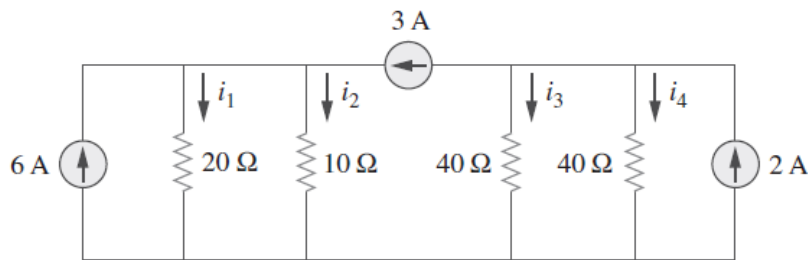
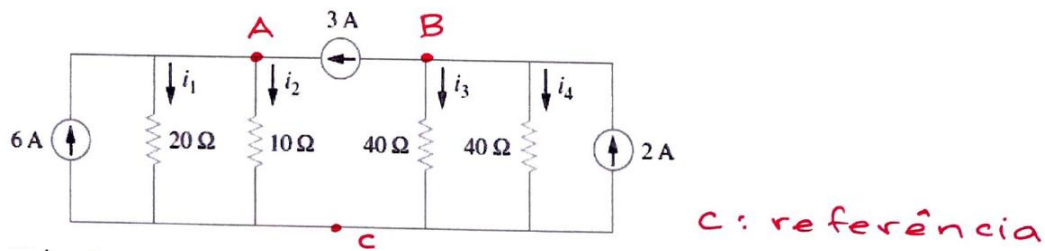


Figura 3.53 Esquema para o Problema 3.4.

SOLUÇÃO



no' A:

$$i_1 + i_2 = 6 + 3$$

$$\frac{v_A}{20} + \frac{v_A}{10} = 9$$

$$v_A + 2v_A = 180$$

$$3v_A = 180 \Rightarrow v_A = 60V$$

no' B:

$$i_3 + i_4 + 3 = 2$$

$$\frac{v_B}{40} + \frac{v_B}{40} = -1$$

$$\frac{2v_B}{40} = -1 \Rightarrow 2v_B = -40$$

$$\Rightarrow v_B = -20V$$

$$\Rightarrow i_1 = \frac{U_A}{20} \Rightarrow \boxed{i_1 = 3A}$$

$$i_2 = \frac{U_A}{10} \Rightarrow \boxed{i_2 = 6A}$$

$$i_3 = \frac{U_B}{40} \Rightarrow \boxed{i_3 = -0,5A}$$

$$i_4 = \frac{U_B}{40} \Rightarrow \boxed{i_4 = -0,5A}$$

3.5 Obtenha v_o no circuito da Figura 3.54.

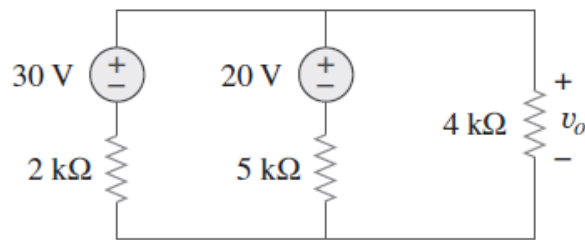
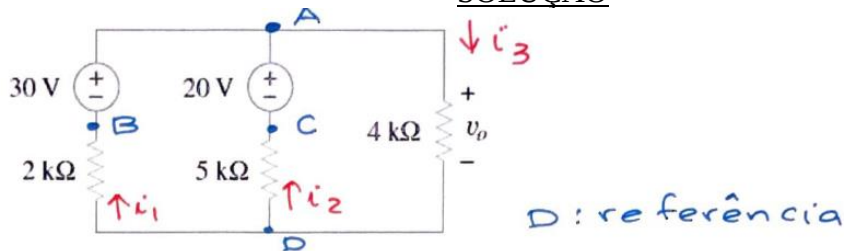


Figura 3.54 Esquema para o Problema 3.5.

SOLUÇÃO



nó A:

$$i_1 + i_2 = i_3$$

$$-\frac{v_B}{2} - \frac{v_C}{5} = \frac{v_A}{4}$$

$$-10v_B - 4v_C = 5v_A$$

$$\Rightarrow \boxed{5v_A + 10v_B + 4v_C = 0}$$

fonte 30V:

$$\boxed{v_A - v_B = 30}$$

fonte 20V:

$$\boxed{v_A - v_C = 20}$$

$$\begin{cases} 5u_A + 10u_B + 4u_C = 0 \\ u_A - u_B = 30 \Rightarrow u_B = u_A - 30 \\ u_A - u_C = 20 \Rightarrow u_C = u_A - 20 \end{cases}$$

$$\Rightarrow 5u_A + 10u_B + 4u_C = 0$$

$$5u_A + 10(u_A - 30) + 4(u_A - 20) = 0$$

$$5u_A + 10u_A - 300 + 4u_A - 80 = 0$$

$$\Rightarrow 19u_A = 380$$

$$u_A = \frac{380}{19} \Rightarrow \boxed{u_A = 20V}$$

$$u_0 = u_A \Rightarrow \boxed{u_0 = 20V}$$

3.6 Use a análise nodal para calcular V_1 no circuito da Figura 3.55.

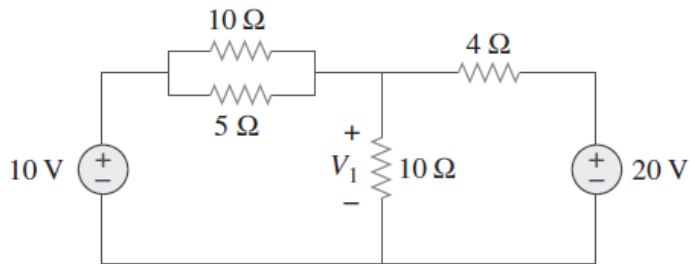
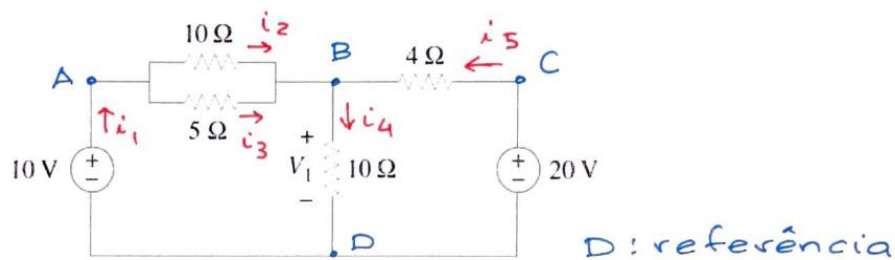


Figura 3.55 Esquema para o Problema 3.6.

SOLUÇÃO



nó A:

$$V_A = 10V$$

nó C:

$$V_C = 20V$$

nó B:

$$i_2 + i_3 + i_5 = i_4$$

$$\frac{V_A - V_B}{10} + \frac{V_A - V_B}{5} + \frac{V_C - V_B}{4} = \frac{V_B}{10}$$

$$2V_A - 2V_B + 4V_A - 4V_B + 5V_C - 5V_B = 2V_B$$

$$6V_A + 5V_C = 13V_B$$

$$\Rightarrow 13V_B = 6 \times 10 + 5 \times 20$$

$$13V_B = 160$$

$$\Rightarrow U_B = \frac{160}{13} \Rightarrow \boxed{U_B = 12,31 \text{ V}}$$

$$U_1 = U_B \Rightarrow \boxed{U_1 = 12,31 \text{ V}}$$

3.11 Determine V_o e a potência dissipada em todos os resistores no circuito da Figura 3.60.

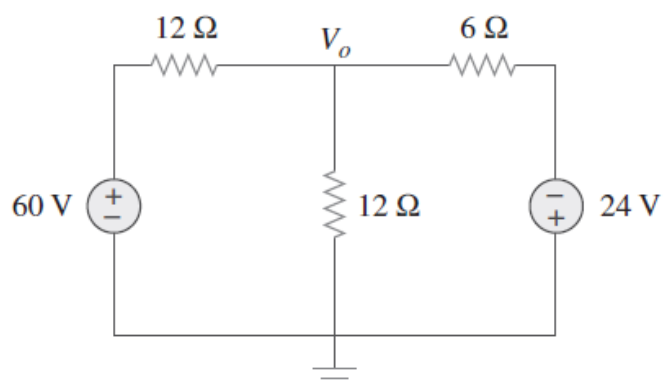
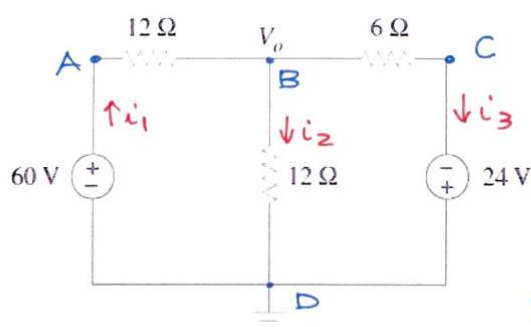


Figura 3.60 Esquema para o Problema 3.11.

SOLUÇÃO



D: referência

no' A:

$$V_A = 60V$$

no' C:

$$V_C = -24V$$

no' B:

$$i_1 = i_2 + i_3$$

$$\frac{V_A - V_B}{12} = \frac{V_B}{12} + \frac{V_B - V_C}{6}$$

$$v_A - v_B = v_B + 2v_B - 2v_C$$

$$4v_B = v_A + 2v_C$$

$$4v_B = 60 + 2 \cdot (-24)$$

$$4v_B = 60 - 48$$

$$4v_B = 12 \Rightarrow \boxed{v_B = 3V}$$

$$v_D = v_B \Rightarrow \boxed{v_D = 3V}$$

tensões nodais:

$$v_A = 60V$$

$$v_C = -24V$$

$$v_B = 3V$$

$$v_A - v_B = 57V$$

$$v_B - v_C = 27V$$

correntes:

$$i_1 = 4,75A$$

$$i_2 = 0,25A$$

$$i_3 = 4,5A$$

potências:

$$R_1 = 12\Omega : P_1 = \frac{57^2}{12} \Rightarrow \boxed{P_1 = 270,75W}$$

$$R_2 = 12\Omega : P_2 = \frac{3^2}{12} \Rightarrow \boxed{P_2 = 0,75W}$$

$$R_3 = 6\Omega : P_3 = \frac{27^2}{6} \Rightarrow \boxed{P_3 = 121,5W}$$

$$\text{fonte } 60\text{V} : P_{f1} = 60 \cdot i_1 = 60 \times 4,75$$

$$\Rightarrow P_{f1} = 285 \text{ W}$$

$$\text{fonte } 24\text{V} : P_{f2} = 24 \cdot i_3 = 24 \times 4,5$$

$$\Rightarrow P_{f2} = 108 \text{ W}$$

*Fonte: ALEXANDER, Charles K.; SADIKU, Matthew N.O. **Fundamentos de Circuitos Elétricos**